



Revealing the arboreal frog genus *Auparoparo*: A new classification after 130 years of *Oreophryne* sensu lato, subfamily Asterophryinae (Anura: Microhylidae)

Journal:	<i>Zoological Journal of the Linnean Society</i>
Manuscript ID	Draft
Manuscript Type:	Original Article
Keywords:	phylogenetic nomenclature < Phylogenetics, morphology < Anatomy, Phylogenetics, homoplasy < Phylogenetics, Indo-Pacific < Geography, frogs < Taxa, new classification < Taxonomy, new genera < Taxonomy, phylogenetic systematics < Phylogenetics, Anura < Taxa
Abstract:	<i>Oreophryne</i> sensu lato (Boettger 1895; 75 species) is the largest genus of the microhylid frog subfamily Asterophryinae (Anura: Microhylidae), contains nearly all its arboreal species, and has a surprisingly expansive distribution across Papua New Guinea to Southeast Asia. Hill et al. (2022) demonstrated that <i>Oreophryne</i> is polyphyletic, consisting of two unrelated reciprocally monophyletic genera. We acquired type species samples (<i>O. senckenbergiana</i>) from North Maluku Island, Indonesia, allowing for a decisive phylogenetic resolution for which clade retains the name <i>Oreophryne</i>. Here we show that the type species is a member of the more basal clade ("Oreophryne A" in Hill et al. 2022) indicating that <i>Oreophryne</i> is the sister clade to the rest of Asterophryinae. The North Maluku specimens are closely related and diverged with New Guinea taxa about 10 MYA. We propose <i>Auparoparo</i> gen. nov. for the other clade ("Oreophryne B"). We employ phylogenetic taxonomy to redefine <i>Oreophryne</i> and <i>Auparoparo</i> and show that they are morphologically undiagnosable. Our findings clearly demonstrate that a second clade of primarily arboreal frogs has evolved within Asterophryne, making these two clades strong candidates for convergent evolution. More complete sampling of additional species with greater range can help to further inform the dynamics of ecological diversification.

Abstract

Oreophryne sensu lato (Boettger 1895; 75 species) is the largest genus of the microhylid frog subfamily Asterophryinae (Anura: Microhylidae), contains nearly all its arboreal species, and has a surprisingly expansive distribution across Papua New Guinea to Southeast Asia. Hill et al. (2022) demonstrated that *Oreophryne* is polyphyletic, consisting of two unrelated reciprocally monophyletic genera. We acquired type species samples (*O. senckenbergiana*) from North Maluku Island, Indonesia, allowing for a decisive phylogenetic resolution for which clade retains the name *Oreophryne*. Here we show that the type species is a member of the more basal clade (“*Oreophryne* A” in Hill et al. 2022) indicating that *Oreophryne* is the sister clade to the rest of Asterophryinae. The North Maluku specimens are closely related and diverged with New Guinea taxa about 10 MYA. We propose ***Auparoparo* gen. nov.** for the other clade (“*Oreophryne* B”). We employ phylogenetic taxonomy to redefine *Oreophryne* and *Auparoparo* and show that they are morphologically undiagnosable. Our findings clearly demonstrate that a second clade of primarily arboreal frogs has evolved within Asterophryne, making these two clades strong candidates for convergent evolution. More complete sampling of additional species with greater range can help to further inform the dynamics of ecological diversification.

Keywords : frogs; homoplasy; indo-pacific; morphology; new classification; new genera; phylogenetic nomenclature; phylogenetics

Introduction

The frog genus *Oreophryne* sensu lato (Boettger 1895) is the largest genus of the hyperdiverse Asterophryinae, the largest subfamily of Microhylidae. With 75 diagnosed species, *Oreophryne* sensu lato makes up one fifth of the currently ~365 recognized species of Asterophryinae (Hill et al. 2022; Frost 2024), despite being only one of ~16 genera. *Oreophryne* sensu lato are conspicuously arboreal (or scansorial) and the genus accounts for nearly all arboreal species of Asterophryinae (Günther et al. 2001; Menzies 2006). *Oreophryne* sensu lato also has an unusually large geographic distribution, the largest of any genus in its subfamily (Hill et al. 2023a), with species centered on mainland New Guinea (Zweifel et al. 2003; Frost 2024) and its offshore islands including the D'Entrecasteaux and Louisiade Archipelagos (Zweifel 1956; Kraus and Allison 2004; Kraus and Allison 2009; Kraus 2013), New Britain Island (Werner 1898; Zweifel et al. 2003), but ranging westward (Müller 1894; Boulenger 1896; Ahl 1933; Putri et al. 2023) and northward to Indonesia (Boettger 1895; Setiadi et al. 2010; Günther et al. 2023) and the Southern Philippines Islands (Alcala 1986; Balinton and Seña 2015). Most of species are small, arboreal or scansorial frogs that inhabit lowland

to montane rain forest, but there is a small group of diminutive species that inhabit high elevation grasslands, some of which are terrestrial (Zweifel et al. 2005; Günther and Richards 2012; Putri et al. 2023).

Although the generic taxonomy of the subfamily Asterophryinae has had a turbulent history (reviewed in Hill et al. 2022), the taxonomy of the genus *Oreophryne* sensu lato in particular has been comparatively stable until recently. Generic designation to *Oreophryne* sensu lato has traditionally relied on a pectoral anatomy that is distinct from all other genera, with highly reduced and distinctly curved clavicles (Parker 1934; Zweifel 2003). However, this trait has never been demonstrated to be a synapomorphy. Hill et. al (2022, 2023b) established that *Oreophryne* sensu lato represents two reciprocally monophyletic groups, temporarily labeled “*Oreophryne* A” and “*Oreophryne* B”. Because the type species *Oreophryne senckenbergiana* has not yet been included in large genus-level molecular phylogenies, it has not been possible to decipher which of the two groups can retain the genus name *Oreophryne*.

Taxonomic History:

Boettger (1895) described the genus *Oreophryne* sensu lato and the species *Oreophryne senckenbergiana* on the basis of two specimens from Halmahera Island, in the Maluku Archipelago west of New Guinea, that lacked procoracoids and had cartilaginous sterna. He also noted the presence of expanded finger pads and transverse folds of skin in the palate. Parker (1934) noted that procoracoids were

1
2
3 present, but were reduced and re-diagnosed the genus on the basis of this feature
4
5 together with the presence of reduced clavicles, no omosternum and a large
6
7 cartilaginous sternum. Zweifel (2003) noted that “the clavicles are tiny, slightly curved
8
9 bones lying on (ventral to) the procoracoid cartilage and apart from the midline and the
10
11 scapula.” On the basis of these features there are now 75 recognized species in the
12
13
14
15 genus (Frost 2024).
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56

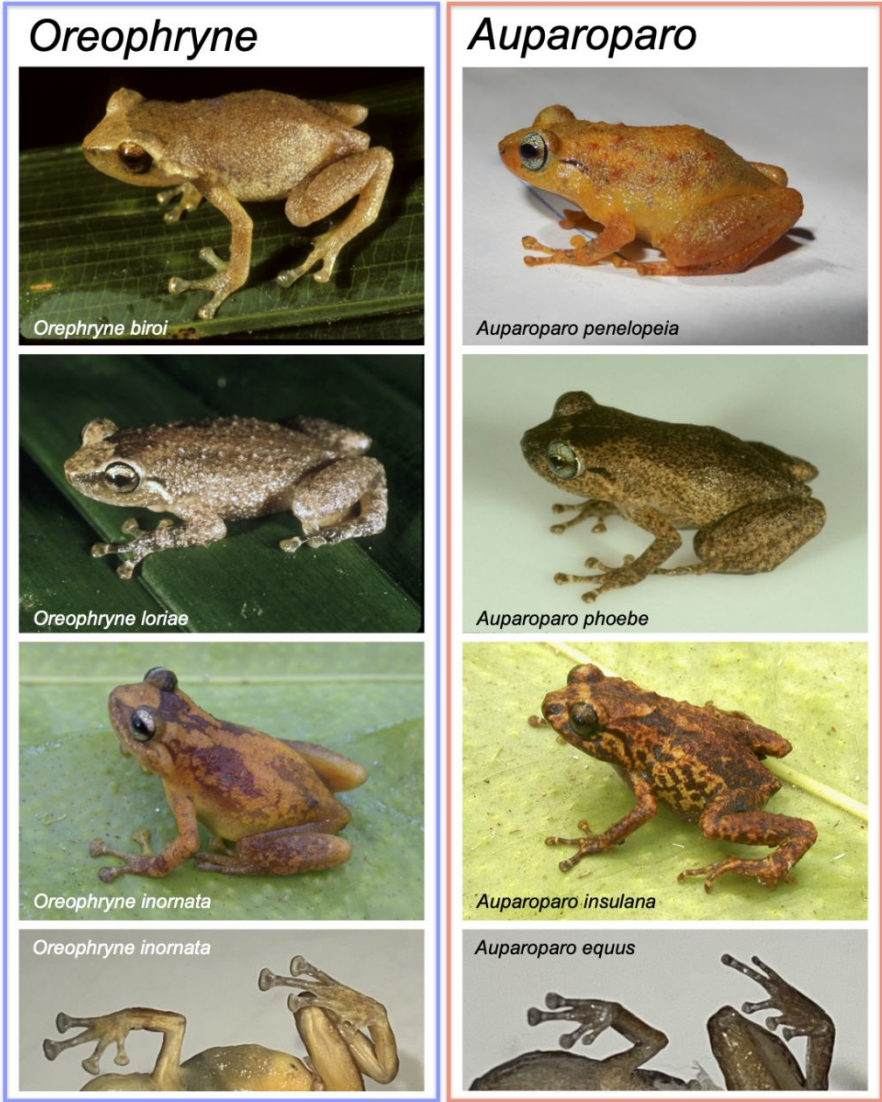


Figure 1: Portraits in life of arboreal frogs in Asterophryinae. *Oreophryne* species on the left, *Auparoparo* on the right, species names as labeled. Comparison of finger and toe pad sizes in *Oreophryne inornata* BPBM16217 vs. *Auparoparo equus* BPBM15775, bottom row.

While *Oreophryne* sensu lato is united morphologically ([Figure 1](#)), large scale molecular phylogenetic analysis with many ingroup samples were required to clarify that this group contains two reciprocally monophyletic clades (Hill et al. 2022, 2023b). Early molecular phylogenetic studies suggested monophyly but only included a few taxa within a larger-scale study (Köhler and Günther 2008 included 9 *Oreophryne* taxa; de Sa et al. 2012 included 4). Rivera et al. (2017) provided the first molecular evidence of a polyphyletic *Oreophryne* sensu lato with 36 *Oreophryne* within a larger phylogeny Asterophryinae which included 155 samples, followed by Tu et al. (2018)'s sparse supermatrix approach with 18 taxa of *Oreophryne* falling out in various places within a much larger phylogeny, but both of these lacked strong nodal support connecting the hypothesized clades to the backbone of the phylogeny. The polyphyletic nature of *Oreophryne* sensu lato was not clearly demonstrated until Hill et. al (2022, 2023b) included 44 taxa of *Oreophryne* with complete data sampling within a 218 taxon phylogeny of Asterophryinae. This phylogeny is the most complete and robust phylogeny to date. The two clades, labeled “*Oreophryne* A” and “*Oreophryne* B” have deep histories of ~ 18MY and ~16MY, respectively, with “*Oreophryne* A” most basal to the rest of the subfamily.

The aims of this study are to assess the phylogenetic position of the type species of *Oreophryne* (*Oreophryne senkenbergiana*) and its close sister taxon (*Oreophryne frontifasciata*) into a large Asterophryinae (subfamily-level) phylogeny, to develop a phylogenetic definition of *Oreophryne* and the new genus, and to name and describe

the new genus. Furthermore, we analyze the morphological characteristics and distribution of these genera to assist collectors when identifying putative *Oreophryne* in the field, and discuss the evolutionary patterns and taxonomic utility of these morphological characters.

Materials and Methods

Taxon Sampling

Asterophryinae is an adaptive radiation with short branch lengths along the backbone separating generic-level taxa (Rivera et al. 2017; Hill et al. 2022, 2023b), we reconstruct the phylogeny of *Oreophryne* sensu lato within the larger context of the subfamily. We include 30 species and 18 unnamed taxa of *Oreophryne* sensu lato within a larger dataset of 235 samples of 204 taxa of Asterophryinae collected across New Guinea, the Philippines, and Indonesia with three outgroup species to root the phylogeny (*Dyscophus antongilii*, *Scaphiophryne marmorata*, and *Platypelis grandis*). Importantly, we add the type species *Oreophryne senckenbergiana* (Boettger 1895) = *O. moluccensis* (Peters and Doria 1878) and the closely related *Oreophryne frontifasciata*, in a molecular phylogeny for the first time. Both taxa were collected in Halmahera, North Maluku Province, Indonesia by Iqbal Setiadi , [Table 1](#), near the type locality for *O.*

1
2
3 *senkenbergiana* (Halmahera, Ternate, and Bacan Islands, Indonesia as documented by
4
5 Peters and Doria 1878; Boettger 1895).
6
7

8 9 Molecular Sequence

10
11
12 We obtained nearly complete DNA sequence data for five loci (three nuclear: Seventh in
13
14 Absentia [SIA], Brain Derived Neurotrophic Factor [BDNF], Sodium Calcium Exchange
15
16 subunit-1 [NXC-1], and two mitochondrial loci (Cytochrome oxidase b [CYTB], and
17
18 NADH dehydrogenase subunit 4 [ND4] from four samples of two taxa not previously
19
20 sequenced and combined these with previously published data (Rivera et al. 2017; Hill
21
22 et al. 2022, 2023b) for an alignment of 240 samples with 207 taxa and 2475 base pairs
23
24 that is 96% complete. All loci were complete for the added samples except for CYTB
25
26 from *Oreophryne frontifasciata*. Sequence quality was evaluated using chromatogram
27
28 quality scores, checks for stop codons indicating pseudogenes, BLAST search (Ye et al.
29
30 2012), and concordance of pairwise similarity with other Asterophryinae sequences.
31
32
33 Primer sequences and detailed methods are provided in (Hill et al. 2023b). We note that
34
35 we excluded from the previous alignment (Hill et al. 2022, 2023b) two samples of
36
37 *Aphantophryne pansa* (BPBM 5299 and BPBM 8312) which were deemed unreliable as
38
39 they came from formalin-fixed material, and three samples that should be confirmed
40
41 with new material (BPBM 22708 *Copiula tyleri*, BPBM 38939 *Copiula* sp. 7, BPBM
42
43 37753 *Paedophryne dekot*). We note that genetic material from 46 species of
44
45 *Oreophryne* sensu lato was not available for this molecular diagnosis.
46
47
48
49
50
51
52
53
54
55
56

Morphology, Behavior, and Geographic Distribution

We collected new morphological characters and reexamined traits compiled from the diagnosing literature , [character states defined in 2](#) for all named species of *Oreophryne* in our phylogenetic dataset. Our sample included all species available in the collections of the Bishop Museum, Honolulu, Hawaii (BPBM): thirteen species of “Oreophryne A”, eight species “Oreophryne B” ([Table 3](#)). Finger and toe pads were photographed and length and width of the toe tips were measured using ImageJ software. Trait information for the remaining species were gathered from the literature, six species of “Oreophryne A” and one “Oreophryne B”. In addition, we compiled information on call type from the literature and collection location from the respective collectors or the BPBM database (global positioning coordinates [GPS]; [Table 3](#)). We examined the ability of morphological and behavioral characters to provide a diagnosis to the genus-level clade consistent with molecular phylogenetics.

Nomenclature

We conducted a comprehensive taxonomic literature review of diagnoses for all species known to belong to the clades “Oreophryne A” and “Oreophryne B” (Hill et al. 2022). We applied the ICZN code (Zoological Nomenclature et al. 1999) as well as the Phylocode (de Queiroz and Cantino 2020). We support the nascent effort to assign indigenous

names in taxonomy whenever possible (e.g., Gillman and Wright 2020). One of us (B. Ilova) is a local indigenous language expert, and provided guidance in proposing the novel genus name, *Auparoparo*, meaning “tree frog” in Motu, a regional language, for *Oreophryne* “B”.

Phylogenetic Reconstruction

We used Partitionfinder in IQTREE2 to find the best-fit evolutionary model followed by Maximum Likelihood (ML) phylogenetic reconstruction (Lanfear et al. 2012; Minh et al. 2020). We tested models partitioned by both locus and codon versus by loci only (15 vs. 5 partition scheme), and found that the locus and codon model was superior by the Akaike Information Criterion. Nodal support values for the ML tree were inferred using 2000 bootstrap replicates.

Phylogenetic Definition

We use a maximum crown-clade definition under the PhyloCode (de Queiroz and Gauthier 1990, 1992; de Queiroz and Cantino 2020), specifically, the largest crown clade originating with the most recent common ancestor of the reference species for the clade (e.g., the type species of the genus), and all extant species of the clade more recently than they do with reference species of other clades (i.e., in this study, other generic-level clades). This definition is robust to uncertainty in basal relationships such

as in the case of adaptive radiations with short branch lengths along the backbone as in Asterophryinae, as well as being robust to identification of new species and new clades.

Analyses, Visualization, and Data Availability

All analyses were conducted and graphics produced in the R Statistical Computing Environment unless noted below (R Core Team 2024). Initial DNA sequence alignment was conducted using MUSCLE (Edgar 2004) with manual adjustments in Mesquite (Maddison and Maddison 2023). Phylogenetic reconstruction was conducted using the IQTREE2 suite of software (Lanfear et al. 2012; Minh et al. 2020).

The known ranges of *Oreophryne* sensu lato were downloaded as shape files from the IUCN (2025) redlist database, and sites from collection locations for *Oreophryne* and *Auparoparo* were mapped using the `sf` and `ggplot2` packages (Wickham 2016; Pebesma 2018), overlaid on a basemap created using `rnaturalearth`, `grid`, and `ggspatial` packages (Massicotte and South 2023; Dunnington 2023). correspondence of the phylogenies to geographic map locations with the `phylo.to.map` function, both from the `phytools` package (Revell 2012). Tree graphics were produced with the `ggtree` package Yu (2020) supplemented by `ggplot2` (Wickham 2016). All data, alignments, and code to reproduce analyses are available in the public repository ...

Results

Phylogeny and Phylogenetic Taxonomy

A partition scheme including codon as well as locus resulted in an improvement of fit by 3,127 AIC units (evolutionary model partitioned by locus and codon, 15-partitions: AIC of 155,492, versus partitioned by locus only, 5 partitions: AIC of 158,619). Unless otherwise noted, all phylogenetic reconstructions used the 15-partition scheme identified as best-fit by IQTREE2.

We recovered a maximum likelihood topology that confirms that *Oreophryne* sensu lato is polyphyletic, with species falling into two independently-evolved, reciprocally monophyletic, highly-supported clades [Figure 2](#). Nearly all nodes are moderately or highly supported, with most of the weakly supported nodes falling along the backbone where there are very short branch lengths as characteristic of an adaptive radiation.

Furthermore, these findings demonstrate that the clade descended from node A should retain the converted clade name (CCN) *Oreophryne*, as the type species, *O.*

senckenbergiana, as well as a closely-related sister species *O. frontifasciata*, both from North Maluku Island, Indonesia, are situated within *Oreophryne* with high support ([Figure 2](#)). We propose the new clade name (NCN) *Auparoparo* for the node descended from node B. *Oreophryne* is the oldest clade in the Papuan subfamily, with genera

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56

Cophixalus and *Paedophryne* arising before *Auparoparo*, which split from the remaining genera of Asterophryinae.

For Review Only

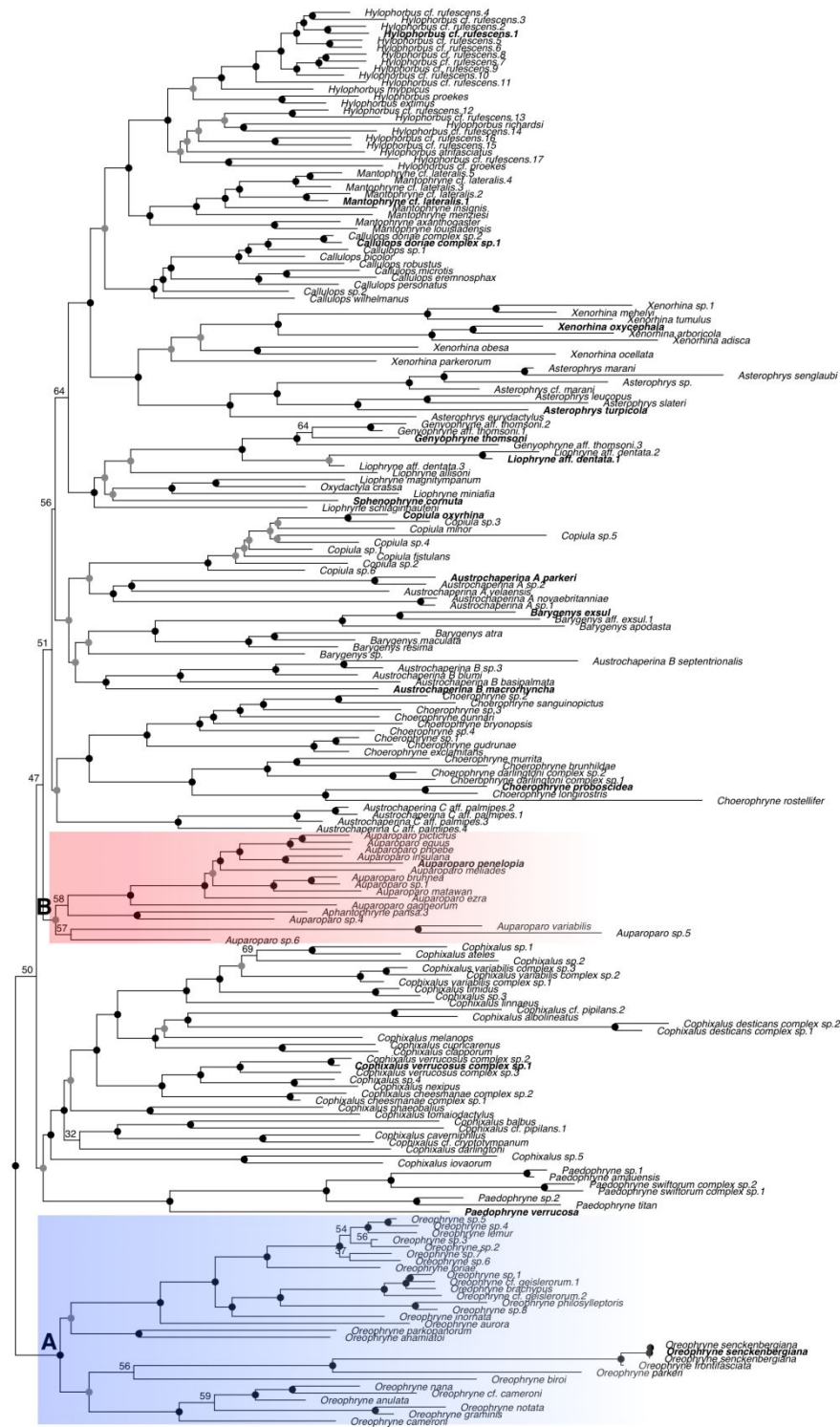


Figure 2: Molecular phylogenetic hypothesis of Asterophryinae showing polyphyly of *Oreophryne* sensu lato, into two reciprocally monophyletic clades, node A is *Oreophryne* (blue clade) and node B is *Auparoparo* (red clade) based on the position of the type species *Oreophryne senkenbergiana*. Molecular evidence includes two mitochondrial and three nuclear markers analyzed using Maximum Likelihood following Rivera et al. (2017); Hill et al. (2022). Nodal support: black dots indicate ML bootstrap support $\geq 95\%$, grey dots $\geq 75\%$, numeric values given for nodes below 75% support. Outgroup taxa excluded.

Morphology, Behavior, and Ranked Taxonomy

We find no morphological synapomorphies that distinguish *Auparoparo* from *Oreophryne*. *Oreophryne* sensu lato is diagnosed by characters of the pectoral girdle: possessing highly reduced, curved procoracoid cartilages and clavicles, and lacking an omosternum (Parker 1934; Zweifel 2003). We found that this suite of characters are present in both *Oreophryne* and *Auparoparo*, and therefore is not diagnostic for neither *Oreophryne* nor *Auparoparo* (Table 3).

Two traits imperfectly segregate between the clades, toe pad widths and call type (Table 3), which may assist in preliminary identification. *Oreophryne* is an arboreal clade with expanded digital pads of roughly equal width between the fingers and toes, whereas *Auparoparo* have visibly larger finger pads, with finger pads roughly 20-30% wider than toes. Both *Oreophryne* and *Auparoparo* use a variety of call types, with “honks” or “peeps” used only by *Oreophryne*, whereas “whinny” calls are used only by

arboreal *Auparoparo*. A minority of species in both groups use the “rattle” call type. Both toe pad width and call type are variable traits with some overlap and thus not reliable for diagnosis. Furthermore, toe pad is clearly an adaptive trait for arboreality

.Geographic Distribution

The geographic distributions of *Auparoparo* and *Oreophryne* are shown in [Figure 3](#), with corresponding phylogenetic relationships.

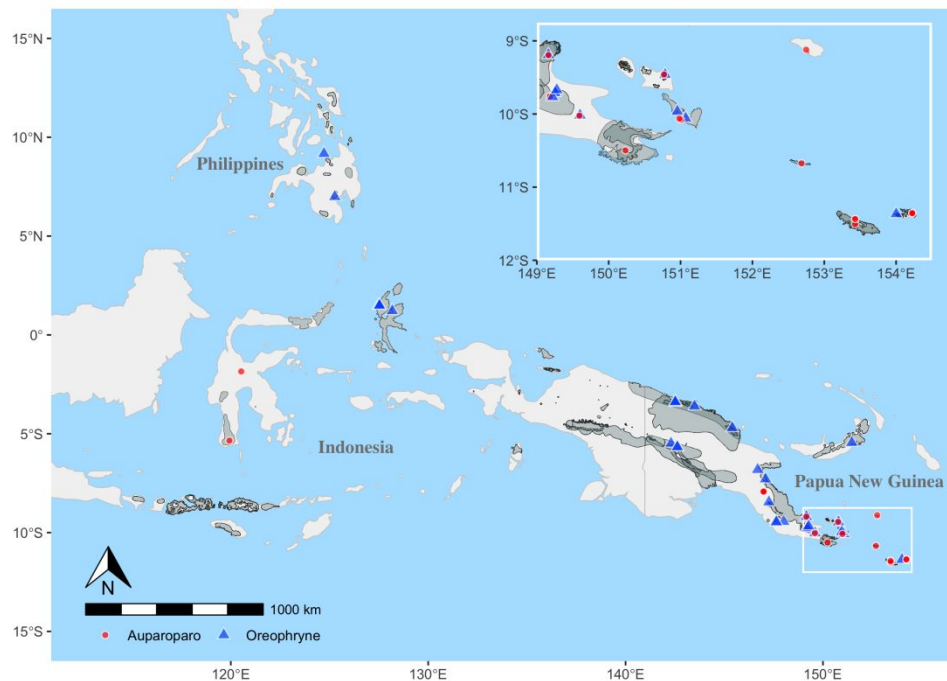


Figure 3: The known distribution of *Oreophryne sensu lato* is plotted in grey polygons (data source IUCN 2025), with points indicating

Auparoparo (red) and *Oreophryne* (blue) as identified by our phylogeny
Figure 2. Note that there is overlap among points

Taxonomy

We use a maximum crown-clade definition for designating *Oreophryne* and *Auparoparo* because the monophyly of these taxa are strongly supported, and the relationships of these taxa to each other and to sister genera of Asterophryinae are well-supported (this study, Hill et al. 2022, 2023b), as has been done for other anuran taxa (Brown et al. 2015) and summarized in the PhyloCode (de Queiroz and Gauthier 1990, 1992, 1994).

That is, we adopt a nodal definition of the clade that identifies the genus as a monophyletic clade. Specifically, to include the type species of the genus and the most recent common ancestor that includes all extant species of the genus but not members of other genera. This definition is robust to uncertainty in basal relationships such as in the case of adaptive radiations with short branch lengths along the backbone as in Asterophryinae, as well as being robust to identification of new species. We follow Brown et al. (2015) in using the terms converted clade name (CCN) to indicate the phylogenetic redefinition of Oreophryne, and new clade name (NCN) for newly designated names (*Auparoparo*), whereas the previously defined polyphyletic *Oreophryne* is indicated by “sensu lato”.

Family Microhylidae (Günther 1858)

Subfamily Asterophryinae (Günther 1858)

Genus *Oreophryne* (Boettger 1895) amended

Microhyla (Peters and Doria 1878)

Callula

Oreophryne (Boettger 1895) Type species: *Oreophryne senckenbergiana*

(Boettger 1895) (= *Microhyla achatina* var. *moluccensis* (Peters and Doria 1878), by morphological monotypy. Placed on Official List of Generic Names in Zoology by Opinion 1266, Anonymous, 1984.

Sphenophryne

Phrynixalus

Mehelyia (Wandolleck 1911), Type species: *Meheylia lineata* (Wandolleck 1911) (= *Sphenophryne biroi* (Méhely 1897), by subsequent designation of (Parker 1934).
Synonymy by (Nieden 1926).

Cophixalus

Chaperina

Type species *Oreophryne senckenbergiana* (Boettger 1895)

Diagnosis: We note that there are no synapomorphies in adult *Oreophryne* that differ from the morphologically similar genus *Auparoparo*. Diagnoses should be made by molecular phylogenetic analysis, based on the phylogenetic definition below.

Description: *Oreophryne* differs from most other genera in subfamily Asterophryinae, but not from *Auparoparo*, by the combination of (1) presence of highly reduced clavicles (vs absence or elongate) which have a distinctly curved shape; (2) presence of well developed digital pads, often with finger and toe pads of similar size; (3) arboreal lifestyle (perch height >2m above ground); (4) most species range in body size from 16mm (*O. graminis*, Günther and Richards 2012) to 30mm, with a few larger species ~40mm (*O. inornata*); (5) small to invisible tympanum; (6) large eyes with pupil horizontal; (7) occasional presence of toe webbing; (8) thin, elongated limbs; and (9) in many species males that call with “honk” or “peep” types or in some species “rattle” type. Some species of *Oreophryne* may be distinguished from *Auparoparo* by roughly

equal finger to toe pad ratio and males calling with “honk” or “peep” but not “whinny” call types.

Phylogenetic definition: *Oreophryne* (CCN) is defined as the maximum crown-clade (the monophyletic clade descending from node A, [Figure 2](#)) originating with the most recent common ancestor of *Oreophryne senckenbergiana* (Boettger 1895), the type species of *Oreophryne*, and all extant species that share a common ancestor with *O. senckenbergiana* more recently than they do with reference species of all the other nominal generic clades listed in [Table 4](#); with reference species of the closest sister clades to *Oreophryne* currently being *Auparoparo penelopeia* (Kraus 2016), *Cophixalus verrucosus* (Boettger 1892), and *Paedophryne kathismaphlox* (Kraus 2010).

Content: Our results support the retention of 18 species in *Oreophryne* with molecular evidence ([Table 5](#)): *Oreophryne anamiatoi* (Kraus and Allison 2009), *O. anulata* (Stejneger 1908), *Oreophryne aurora* (Kraus 2016), *Oreophryne biroi* (Méhely 1897), *Oreophryne brachypus* (Werner 1898), *Oreophryne cameroni* (Kraus 2013), *Oreophryne frontifasciata* (Horst 1883), *Oreophryne geislerorum* (Boettger 1892), *Oreophryne graminis* [Günther and Richards (2012)], *Oreophryne inornata* (Zweifel 1956), *Oreophryne lemur* (Kraus 2016), *Oreophryne loriae* (Boulenger 1898), *Oreophryne nana* (Brown and Alcala 1967), *Oreophryne notata* (Zweifel 2003), *Oreophryne parkeri* (Loveridge 1955), *Oreophryne parkopanorum* (Kraus 2013), *Oreophryne philosylleptoris* (Kraus 2016), and *Oreophryne senckenbergiana* (Boettger

1895). A number of species were not available for our study but should molecular evidence demonstrate their relationship within *Oreophryne*, those species should also retain their generic identity.

Distribution: The range of confirmed species in this dataset remains nearly as broad as previously defined (Figure 3), from the Philippines and Indonesia to the southeastern most tip of mainland New Guinea and several near-shore islands (Fergusson, and Normanby Is.) and the distant Rossel Island of the Louisiade Archipelago. However, the transfer of species which occupy Woodlark Islands and Misima and Sudest Islands of the Louisiade Archipelago reduces the known range of *Oreophryne*. Woodlark Island is inhabited by *O. phoebe* [= *Auparoparo phoebe*], Misima Island is inhabited by *O. picticus* [= *Auparoparo picticus*], and Sudest Island is inhabited by *O. ezra* and *O. meliades* [= *Auparoparo ezra* and *Auparoparo meliades*].

Family Microhylidae (Günther 1858)

Subfamily Asterophryinae (Günther 1858)

Genus *Auparoparo* Nov.

Oreophryne (Boettger 1895)

Sphenophryne (anthonyi) (Boulenger 1897)

Type species *Oreophryne penelopeia* (Kraus 2016) [= *Auparoparo penelopeia*]

Diagnosis: We note that there are no synapomorphies in adult *Auparoparo* that differ from the morphologically similar genus *Oreophryne*. Diagnoses should be made by molecular phylogenetic analysis, based on the phylogenetic definition below.

Description: *Auparoparo* differs from most other genera in subfamily Asterophryinae, but not from *Oreophryne*, by the combination of (1) presence of highly reduced clavicles (vs absence or elongate) which have a distinctly curved shape; (2) presence of well developed digital discs, with finger discs notably wider than toe discs (typically 20% or more); (3) arboreal lifestyle (perch height >2m above ground); (4) small to moderately large body size in known species, from 16mm (*A. brunnea*, Kraus 2017) to 25 mm (*A. picticrus*, Kraus 2016); (5) small to invisible tympanum; (6) large eyes with pupil horizontal; (7) occasional presence of toe webbing; (8) and thin, elongated limbs; and (9) and in many species males that call with a “whinny” call type or in some species “rattle” type. Some species of *Auparoparo* may be distinguished from *Oreophryne* by larger finger pads than toe pads and males calling with “whinny” but not “honk” or “peep” call types.

Phylogenetic definition: *Auparoparo* (NCN) applies to the largest crown clade originating in (CCN) is defined as the maximum crown-clade originating with the most recent common ancestor of *Auparoparo penelopeia* (Kraus 2016), the type species of *Auparoparo*, and all extant species that share a common ancestor with *A. penelopeia* more recently than they do with the type species of all the other nominal genera listed in

[Table 4](#), the type species of the closest sister taxa being *Oreophryne senckenbergiana* (Boettger 1895), *Cophixalus verrucosus* (Boettger 1892), and *Paedophryne kathismaphlox* (Kraus 2010).

Content: We transfer eleven named species from *Oreophryne* into *Auparoparo* ([Table 5](#)): *Auparoparo brunnea* (Kraus 2017), *Auparoparo equus* (Kraus 2016), *Auparoparo ezra* (Kraus and Allison 2009), *Auparoparo insulana* (Zweifel 1956), *Auparoparo matawan* (Kraus 2016), *Auparoparo meliades* (Kraus 2016), *Auparoparo penelopeia* (Kraus 2016), *Auparoparo phoebe* (Kraus 2017), *Auparoparo picticus* (Kraus 2016), *Auparoparo nana*, *Auparoparo penelopia* (Kraus 2016), *Auparoparo variabilis* (Boulenger 1896).

Distribution: Southeastern tip of mainland Papua New Guinea and satellite islands (Fergusson, Misima, Normanby, Rossel, Sudest, Woodlark Is.), Sulawesi.

Etymology: The generic name *Auparoparo* comes from the indigenous language, Motu, which is widely spoken in the region of Papua New Guinea in which this genus is found. It is a combination of the words “au” (tree) and “paro paro” (frog). Motu is not a gendered language, however, for nomenclatural purposes we designate the genus *Auparoparo* as feminine, thereby preserving the gender of existing species names.

Discussion

Our phylogenetic results (Hill et al. 2022; 2023b, this study) provide clarity on the classification of the problematic genus *Oreophryne* sensu lato. We confirm that *Oreophryne* sensu lato is polyphyletic and consists of two reciprocally monophyletic clades, “*Oreophryne* A” and “B” (of Hill et al. 2022), originating 18 and 16 MYA, respectively ([Figure 2](#)). Here we designate “*Oreophryne* A” as retaining the name *Oreophryne* based on its containing the type species *Oreophryne senckenbergiana* (Boettger 1895) with high support. *Oreophryne* is the earliest diverged genus in subfamily Asterophryinae, sister to all the other genera. The second clade, “*Oreophryne* B”, is highly supported as a separate, unrelated genus, which contains no known species with historical generic priority. Thus, we designate this genus as *Auparoparo*, a nominal tribute to the Motu language of the indigenous peoples of the region.

The Difficulty with Cryptic Genera

Perhaps the greatest testament to the crypticism of *Oreophryne* (CCN) vs. *Auparoparo* (NCN) is that no taxonomist was able to distinguish that they are separate clades, independently evolved, in the 130 years between their discovery and phylogenetic classification [Boettger (1895); this study]. These species are indeed morphologically very similar ([Figure 1](#)) and our results provide some clues as to why *Oreophryne* taxonomy has been so difficult. We found no morphological synapomorphy for neither

Auparaparo nor *Oreophryne*, and moreover, no morphological feature that can diagnose either clade or both in combination.

The traditional basis for generic designation of species to *Oreophryne* has primarily relied on a signature pectoral anatomy (whether original or revised diagnosis, Parker 1934; Zweifel 2003). Our phylogenetic results clearly show that this trait appears more than once, in both *Oreophryne* and *Auparaparo*, and rather than being a highly conserved trait, it is either a symplesiomorphy (shared ancestral character) or a result of convergent or parallel evolution, and perhaps adaptive for arboreality. Thus, the pectoral anatomy is not a reliable diagnostic character in differentiating between *Oreophryne* and *Auparaparo*. In studies across a wide diversity of anurans, Soliz et al. (2025) found that anatomical variation of the pectoral girdle corresponds with functional and ecological lifestyle demands, whereas Engelkes et al. (2020) found strong phylogenetic influence in the shape of the pectoral girdle. This point would be excellent for further study within Asterophryinae, as there are high elevation terrestrial species of *Oreophryne* sensu lato that possess the same pectoral trait (Zweifel et al. 2005), as well as a few arboreal species within other genera (*Cophixalus*, *Asterophrys*, *Xenorhina*) with unknown pectoral morphology.

We further reexamined additional characters previously used to diagnose *Oreophryne* as well as characters conventionally used to diagnose species of frogs. We conclude that pectoral girdle connector type (Parker 1934) and relative lengths of the 3rd and 5th toe

are uninformative and the degree of toe webbing is subjective. It was previously suggested that species in *Oreophryne* sensu lato could be divided into two groups based on the type of connection between the procoracoid cartilages and the scapula (Kraus 2013), with roughly half possessing a cartilaginous rod extending to the scapula and others a ligamentous connection (Parker 1934). However, our phylogenetic evidence shows that these traits readily evolve and are not synapmorphies and therefore not reliable indicators of monophyletic groups. Ligamentous connectors are common across both genera ([Table 3](#)). Of species with toe webbing, *Oreophryne* tends to be described as having “well webbed” toes compared to *Auparoparo*’s “basal” webbing ([Table 3](#)). However, intra-specific variability combined with the subjective nature of these descriptions and inconsistency between specific markers when used (“reaching the penultimate tubercle”, “webbed 1/3 length of 5th toe”, etc.), makes this feature difficult to standardize for generic diagnostic use. Many of the morphological traits proposed by Parker (1934) and Zweifel (2003) may in fact be a result of homoplasy to evolutionary pressures imposed by lifestyle or parallel evolution, and while they may prove useful for ecomorphological identification or species diagnoses in combination with other traits, are not informative for generic diagnosis.

We found only two characters with potential for imperfectly identifying genera: relative digital pad size (Parker 1934; Zweifel 1956), and call type (e.g., Kraus 2016). We found that the relative widths of the digit pads differs between genera: *Auparoparo* tends to

1
2
3 have visibly larger finger pads in comparison to toe pads, whereas in *Oreophryne* the
4
5 finger and toe pads are roughly equal in size ([Figure 1](#), bottom row), although there is
6
7 some overlap in finger to toe pad ratio ([Table 3](#)). In some cases, *Auparoparo*'s finger
8
9 pads display T-shapes while the toes are often rounder and narrower. Whether there is
10
11 any performance advantage in larger finger pads compared to toes is unknown.
12
13
14 Furthermore, a handful of high-elevation species not included in our phylogeny have
15
16 evolved terrestrial or fossorial habits and possess smaller digital pads compared to their
17
18 presumed arboreal relatives (Zweifel et al. 2005).
19
20
21
22 Call features are increasingly used for diagnosing species of *Oreophryne*, with
23
24 advertisement call types commonly categorized as rattle, honk, whinny, peeping, or
25
26 clicking (Kraus 2016). We found honk or peep call types are only found in *Oreophryne*
27
28 species in our sample, while whinny calls are only found in *Auparoparo* species, and
29
30 rattle calls are found in both genera ([Table 3](#)). We note that several authors described
31
32 species of each genus, thus the difference between genera cannot be attributed to the
33
34 investigator. However, more complete sampling may add further complexity. As far as
35
36 we are aware there has been no comprehensive acoustic analysis across *Oreophryne*
37
38 sensu lato. More research is needed to explore call diversity both between species and
39
40 across genera to understand call evolution and whether there any features that are
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56 distinctive of genera.

Geographic Distribution

Both *Oreophryne* and *Auparoparo* are geographically centered in the mainland New Guinea. *Oreophryne* extends southward to the Louisiade Archipelago and northward to the South Philippine Islands (Hill et al. 2023b). Addition of the type species *Oreophryne senkenbergiana* and *O. frontifasciata* expands the known range eastward to North Maluku, Indonesia ([Figure 3](#)).

The known distribution of *Auparoparo* is mainly clustered in the southeastern tip of mainland Papua New Guinea (in the East Papuan Composite Terrane) and satellite islands with two species in Sulawesi, Indonesia. This suggests the existence of yet undescribed or undiscovered species of *Auparoparo* with intermediate localities. This geographic range is sure to grow with additional phylogenetic sampling. For example *Oreophryne* (sensu lato) *chlorops* (Günther et al. 2023) is a very large (SVL > 40mm) species recently discovered from mid-elevation mountains of the bird's neck region of Western Papua with clearly enlarged finger pads relative to toe pads. If confirmed as *Auparoparo*, it would significantly expand the range across New Guinea Island.

We found geographic range overlap between *Oreophryne* and *Auparoparo* at multiple localities: on the mainland along the Mt. Dayman Foothills, at the Mt. Simpson Massif, Mt. Trafalgar, and on the islands of Normanby and Fergusson (both nearshore islands), and Rossel island (hundreds of kilometers away), which presents interesting questions

1
2
3 regarding their ecological relationships. For example, *Oreophryne inornata* and
4
5 *Auparoparo insulana* have been collected at the same site (Goodenough Island) and
6
7 even the same microhabitats (Zweifel 1956). How can species coexist in the same
8
9 niche? And what is the temporal history of each colonization? Are both genera also
10
11 present in the nearby islands of Normanby and Fergusson (all part of the
12
13 D'Entrecasteaux Archipelago)? Obtaining additional geographical samples for inclusion
14
15 in phylogenetic analysis should be a high priority to gain a more complete
16
17 understanding of the geographic distribution and evolutionary history of both
18
19 *Oreophryne* and *Auparoparo*.
20
21
22
23

24 25 Phylogenetic Taxonomy 26 27

28 We adopt an explicitly phylogenetic framework for the new taxonomy of the genera
29
30 *Oreophryne* and *Auparoparo*, and to our knowledge, this is the first time that a
31
32 phylogenetic taxonomic approach has been applied for any genus of the subfamily
33
34 Asterophryinae. A nodal definition of genera is advantageous especially because
35
36 Asterophryinae is an adaptive radiation. While we are increasingly confident that at least
37
38 16 genera are indeed monophyletic, nevertheless, very short branch lengths separate
39
40 many of the genera along the backbone [Hill et al. (2022); this study]. A nodal definition
41
42 is robust to uncertainty (i.e., rearrangements of order) along the backbone. We define
43
44 genera as the largest crown clade originating with the most recent common ancestor of
45
46 the reference species for the clade (e.g., the type species *Oreophryne senkenbergiana*),
47
48
49
50
51
52
53
54
55
56

1
2
3 and all extant species of the clade more recently than they do with reference species of
4
5 other clades that the sample may potentially belong to. The type species, if genetic
6
7 material is available seems like a reasonable choice for the reference taxon, but for
8
9 groups such as Asterophryinae, where a number of type species were collected in the
10
11 1800's it may not be possible to obtain a genetic sample, as these locations may be
12
13 uncertain and/or remote, and other proxies may be needed. Nevertheless, is
14
15 increasingly clear that for *Oreophryne* and *Auparoparo*, a molecular phylogenetic
16
17 approach is required for final confirmation of taxonomy.
18
19
20
21

22 Furthermore, a phylogenetic definition provides guidance for robust classification. Given
23
24 the deep histories of the two genera, it is imperative that molecular confirmation of
25
26 either *Oreophryne* or *Auparoparo* include members of the other clade as an outgroup,
27
28 but also members of *Cophixalus*, *Paedophryne*, or other closest sister genera as
29
30 outgroups to clearly resolve whether the sample shares an MRCA more closely with
31
32 *Oreophryne* or *Auparoparo*. The ability to resolve the “boundaries” of the monophyletic
33
34 groups requires it. An additional difficulty here is that within each genus, there are
35
36 subclades with deep histories, so care must be taken to decisively demonstrate
37
38 monophyly. For example, a phylogenetic analysis with only *Oreophryne* sensu lato and
39
40 outgroups to Asterophryinae, will show grouping of *Oreophryne* and *Auparoparo*,
41
42 because of lack of intervening phylogenetic history. Similarly, an analysis with only
43
44 *Oreophryne* sensu lato with a member of one of the younger genera, say, *Hylophorbus*,
45
46
47
48
49
50
51
52
53
54
55
56

will give a similar misleading result because both *Oreophryne* and *Auparoparo* are equally distant from *Hylophorbus*. Clearly a greater taxonomic representation is always better, but in this case, care is required to avoid a misleading result. This study furthermore highlights the importance of taxonomic sampling for robust phylogenetic results. Developing an orderly taxonomy in a clade with over 350 named species (and an estimated 40% more yet to be described species currently in museum collections) would be that much more challenging without the benefit of generic diagnoses in this interesting group.

Clarifying the generic structure also facilitates appreciation for the patterns of diversification: a tendency toward niche conservatism within genera, ecological and generic diversification occurring early in the history of the subfamily (Hill et al. 2022); and furthermore the very interesting biogeographical history of simultaneous dispersal of multiple genera to offshore islands (Hill et al. 2022, 2023a) revealing the very strong influence of plate tectonics for explaining the historical dispersal patterns of this group. Finally, without stability in the naming of genera and the independent histories embodied by each, it becomes much more difficult to appreciate that the two clades represented by *Oreophryne* and *Auparoparo* have independently evolved, that there are two independent arboreal clades, and would make it very difficult to appreciate the impressive convergence or parallelism embodied by these speciose and successful genera. An accurate and informative taxonomy serves to advance our understanding of

1
2
3 this group and the very interesting evolutionary insights that they embody leading to
4
5 further discovery.
6
7

8
9 Lastly, we emphasize that future naming of species, genera, and higher classifications
10
11 should always seek to recognize the language of the indigenous stewards of the land
12
13 and ecosystem from which they were discovered (Gillman and Wright 2020). This is
14
15 especially true of Asterophryinae frogs, whose awe-inspiring diversity mirrors the rich
16
17 cultural beauty and historical complexity of Papuan people.
18
19
20
21
22
23
24
25
26
27
28
29
30

31 Citations

32
33
34
35 Ahl, E. 1933. Ergebnisse der celebes-und halmaheira-expedition heinrich 1930–32.
36
37 Mitteilungen aus dem Zoologischen Museum in Berlin 19:577–583.
38
39

40
41 Alcala, A. C. 1986. [Guide to philippine flora and fauna: Amphibians and reptiles](#). Natural
42
43 Resources Management Center, Ministry of Natural Resources; University of the
44
45 Philippines.
46
47
48
49
50
51
52
53
54
55
56

1
2
3 Balinton, R. B., and C. A. de la Seña. 2015. [Karyotype of a rare minute frog, oreophryne](#)
4 [cf. Anulata \(anura: Microhylidae\), in agusan marsh, mindanao, philippines](#). The
5
6 Herpetological Bulletin.
7
8

9
10
11 Boettger, O. 1892. Katalog der batrachier-sammlung im museum der
12
13 senckenbergischen naturforschenden gesellschaft in frankfurt am main. Frankfurt a. M.:
14
15 Gebrüder Knauer.
16

17
18
19 Boettger, O. 1895. Liste der reptilien und batrachier der insel halmaheira nach den
20
21 sammlungen prof. Dr. W. Kükenthal's. Zoologischer Anzeiger 18:129–138.
22

23
24 Boulenger, G. A. 1898. An account of the reptiles and batrachians collected by dr. L.
25
26 Loria in british new guinea. Annali del Museo civico di storia naturale di Genova, Serie 2
27
28 18:695–710.
29

30
31
32 Boulenger, G. A. 1897. Descriptions of new lizards and frogs from mount victoria, owen
33
34 stanley range, new guinea, collected by mr. A. S. anthony. Annals and Magazine of
35
36 Natural History, Series 6 19:6–13.
37

38
39
40 Boulenger, G. A. 1896. [Descriptions of new reptiles and batrachians obtained by mr.](#)
41 [Alfred everett in celebes and jampea](#). The Annals and magazine of natural history;
42
43 zoology, botany, and geology, Series 6 18:62–64.
44
45
46
47
48
49
50
51
52
53
54
55
56

- 1
2
3 Brown, R. M., C. D. Siler, S. J. Richards, A. C. Diesmos, and D. C. Cannatella. 2015.
4
5 [Multilocus phylogeny and a new classification for southeast asian and melanesian forest](#)
6
7 [frogs \(family ceratobatrachidae\)](#). Zoological Journal of the Linnean Society 174:130–
8
9 168.
10
11
12
13 Brown, Walter C. (Walter Creighton), and Alcala Angel C. 1967. [A new frog of the genus](#)
14
15 [oreophryne and a list of amphibians from camiguin island, philippines](#). Proceedings of
16
17 the Biological Society of Washington 80:65–68. Washington, Biological Society of
18
19 Washington,.
20
21
22
23 de Queiroz, K., and P. Cantino. 2020. [International code of phylogenetic nomenclature](#)
24
25 [\(PhyloCode\)](#). 1st ed. CRC Press.
26
27
28
29 de Queiroz, K., and J. Gauthier. 1992. [Phylogenetic taxonomy](#). Annual Review of
30
31 Ecology, Evolution, and Systematics 23:449–480.
32
33
34 de Queiroz, K., and J. Gauthier. 1990. [Phylogeny as a central principle in taxonomy -](#)
35
36 [phylogenetic definitions of taxon names](#). Systematic Zoology 39:307–322.
37
38
39
40 de Queiroz, K., and J. Gauthier. 1994. [Toward a phylogenetic system of biological](#)
41
42 [nomenclature](#). Trends in Ecology & Evolution 9:27–31.
43
44
45 de Sa, R. O., J. W. Streicher, R. Sekonyela, M. C. Forlani, S. P. Loader, E. Greenbaum,
46
47 S. Richards, and C. F. B. Haddad. 2012. [Molecular phylogeny of microhylid frogs](#)
48
49
50
51
52
53
54
55
56

(Anura: Microhylidae) with emphasis on relationships among New World genera. BMC Evolutionary Biology 12.

Dunnington, D. 2023. [Ggspatial: Spatial data framework for ggplot2](#).

Edgar, R. C. 2004. [Muscle: A multiple sequence alignment method with reduced time and space complexity](#). BMC Bioinformatics 5:113.

Engelkes, K., L. Kath, T. Kleinteich, J. U. Hammel, A. Beerlink, and A. Haas. 2020. [Ecomorphology of the pectoral girdle in anurans \(amphibia, anura\): Shape diversity and biomechanical considerations](#). ECOLOGY AND EVOLUTION 10:11467–11487.

Frost, D. R. 2024. [Amphibian species of the world: An online reference. Version 6.2](#).

Gillman, L. N., and S. D. Wright. 2020. [Restoring indigenous names in taxonomy](#). Communications Biology 3:609.

Günther, A. C. L. G. 1858. On the systematic arrangement of the tailless batrachians and the structure of *rhinophrynus dorsalis*. Proceedings of the Zoological Society of London 339–352.

Günther, R., D. T. Iskandar, and S. J. Richards. 2023. [A new large oreophryne species from the mountains of papua province, indonesian new guinea \(amphibia, anura, microhylidae\)](#). Vertebrate Zoology 73:153–159. Senckenberg Gesellschaft für Naturforschung.

Günther, R., and S. J. Richards. 2012. Five new microhylid frog species from engaprovince, papua new guinea, and remarks on albericus alpestris (anura, microhylidae). VERTEBRATE ZOOLOGY 61:343–372.

Günther, R., S. Richards, and D. T. Iskandar. 2001. Two new species of the genus oreophryne from irian jaya, indonesia (amphibia, anura, microhylidae). Spixiana 24:257–274.

Hill, E. C., C. J. Fraser, D. F. Gao, M. J. Jarman, E. R. Henry, B. Iova, A. Allison, and M. A. Butler. 2022. [Resolving the deep phylogeny: Implications for early adaptive radiation, cryptic, and present-day ecological diversity of papuan microhylid frogs](#). Molecular Phylogenetics and Evolution 177. Academic Press Inc.

Hill, E. C., D. F. Gao, D. A. Polhemus, C. J. Fraser, B. Iova, A. Allison, and M. A. Butler. 2023a. Testing geology with biology: Plate tectonics and the diversification of microhylid frogs in the papuan region. Integrative Organismal Biology.

Hill, E. C., M. J. Jarman, C. J. Fraser, D. F. Gao, E. R. Henry, A. R. Fisher, B. Iova, A. Allison, and M. A. Butler. 2023b. [Molecular and phylogenetic datasets for the asterophryinae frogs of new guinea with additional data on lifestyle, geography, and elevation](#). Data in Brief 47:108987.

Horst, R. 1883. On new and little-known frogs from the malayan archipelago. Notes from the Leyden Museum 5:235–244.

IUCN. 2025. [Asterophryinae \(spatial data\)](#). The IUCN Red List of Threatened Species. [Version 2025-1](#). International Union for Conservation of Nature.

Kampen, P. N. van. 1923. [The amphibia of the indo-australian archipelago](#). E. J. Brill, ltd, Leiden.

Köhler, F., and R. Günther. 2008. [The radiation of microhylid frogs \(amphibia : Anura\) on new guinea: A mitochondrial phylogeny reveals parallel evolution of morphological and life history traits and disproves the current morphology-based classification](#). Molecular Phylogenetics and Evolution 47:353–365. ACADEMIC PRESS INC ELSEVIER SCIENCE, 525 B ST, STE 1900, SAN DIEGO, CA 92101-4495 USA.

Kraus, F. 2017. [A new insular species of oreophryne \(anura, microhylidae\) from papua new guinea](#). Journal of Herpetology 51:552–558. Society for the Study of Amphibians; Reptiles.

Kraus, F. 2010. [New genus of diminutive microhylid frogs from papua new guinea](#). ZOOKEYS 39–59.

Kraus, F. 2016. [Ten new species of oreophryne \(anura, microhylidae\) from papua new guinea](#). Zootaxa 4195:1–68. Magnolia Press.

Kraus, F. 2013. [Three new species of <i>oreophryne</i> \(anura, microhylidae\) from papua new guinea](#). ZOOKEYS 93–121.

Kraus, F., and A. Allison. 2009. [New microhylid frogs from the muller range, papua new guinea](#). ZooKeys 26:53–76.

Kraus, F., and A. Allison. 2004. Notes on poorly known new guinea frogs. I. Microhylids from the d'entrecasteaux islands. HERPETOFAUNA-SYDNEY- 34:26–39.

AUSTRALASIAN AFFILIATION OF HERPETOLOGICAL.

Lanfear, R., B. Calcott, S. Y. W. Ho, and S. Guindon. 2012. [PartitionFinder: Combined Selection of Partitioning Schemes and Substitution Models for Phylogenetic Analyses](#). Molecular Biology and Evolution 29:1695–1701.

Loveridge, A. 1955. New frogs of the genera asterophrys and oreophryne from new guinea. Breviora. Museum of Comparative Zoology, Cambridge, Massachusetts 50:1–5.

Maddison, W. P., and D. R. Maddison. 2023. [Mesquite: A modular system for evolutionary analysis](#).

Massicotte, P., and A. South. 2023. [Rnaturalearth: World map data from natural earth](#).

Méhely, L. 1897. Further contributions to the herpetology of new-guinea. Természetráji Füzetek 20:398–419.

Menzies, J. 2006. The frogs of new guinea and the solomon islands. Pensoft Publishers.

Minh, B. Q., H. A. Schmidt, O. Chernomor, D. Schrempf, M. D. Woodhams, A. von Haeseler, and R. Lanfear. 2020. [IQ-TREE 2: New models and efficient methods for phylogenetic inference in the genomic era](#). *Molecular Biology and Evolution* 37:1530–1534.

Müller, F. 1894. Reptilien und amphibien aus celebes. *Verhandlungen der Naturforschenden Gesellschaft in Basel* 10:825–843.

Nieden, F. 1926. Anura II. engystomatidae. *Das Tierreich* 49:1–110.

Parker, H. W. 1934. A monograph of the frogs of the family microhylidae. London: Trustees of the British Museum.

Pebesma, E. 2018. [Simple Features for R: Standardized Support for Spatial Vector Data](#). *The R Journal* 10:439–446.

Peters, W. C. H., and G. Doria. 1878. *Catalogo dei rettili e dei batraci raccolti da o. Beccari, I.m. D'albertis a.a. Bruijn nella sotto-regione austro-malese*. *Annali del Museo civico di storia naturale di Genova* 13:323–450.

Putri, A., W. TRILAKSONO, H. KURNIATI, A. Hitch, A. Jr, K. Widayati, A. Farajallah, and A. Hamidy. 2023. [A new high elevation species of oreophryne boettger \(anura: Microhylidae\) from sulawesi, indonesia](#). *Zootaxa* 5353:455–467.

- 1
2
3 R Core Team. 2024. [R: A language and environment for statistical computing](#). R
4
5 Foundation for Statistical Computing, Vienna, Austria.
6
7
8
9 Revell, L. J. 2012. [Phytools: An r package for phylogenetic comparative biology \(and](#)
10
11 [other things\)](#). *Methods in Ecology and Evolution* 3:217–223.
12
13
14 Rivera, J. A., F. Kraus, A. Allison, and M. A. Butler. 2017. [Molecular phylogenetics and](#)
15
16 [dating of the problematic new guinea microhylid frogs \(amphibia: Anura\) reveals](#)
17
18 [elevated speciation rates and need for taxonomic reclassification](#). *Molecular*
19
20 *Phylogenetics and Evolution* 112:1–11. Academic Press Inc.
21
22
23
24 Setiadi, M. I., A. Hamidy, Z. Abidin, D. Susanto, R. M. Brown, A. T. Peterson, X. Li, and
25
26 B. J. Evans. 2010. [Genetic structure of herpetofauna on halmahera island, indonesia:](#)
27
28 [Implications for aketajawe-lolobata national park](#). *Conservation Biology* 24:553–562.
29
30
31
32 Soliz, M. C., V. Abdala, and M. J. Tulli. 2025. [The ecological drivers of variation in](#)
33
34 [pectoral girdle anatomy in frogs](#). *Acta Zoologica* 106:38–54.
35
36
37
38 Stejneger, L. 1908. Two new species of toads from the philippines. *Proceedings of the*
39
40 *United States National Museum* 33:573–576.
41
42
43 Tu, N., M. Yang, D. Liang, and P. Zhang. 2018. [A large-scale phylogeny of microhylidae](#)
44
45 [inferred from a combined dataset of 121 genes and 427 taxa](#). *Mol Phylogenet Evol*
46
47 126:85–91.
48
49
50
51
52
53
54
55
56

Wandolleck, Benno. 1911. [Die amphibien und reptilien der papuanischen ausbeute dr. schlaginhaufens](#). Abhandlungen und Berichte des Königl. Zoologischen und

Anthropologisch-Etnographischen Museums zu Dresden 13:1–15. Berlin, R.

Friedlaender, 1887-1917.

Werner, Franz. 1898. [Vorläufige mitteilung über die von herrn prof. F. Dahl im bismarck-archipel gesammelten reptilien und batrachier](#). Zoologischer Anzeiger 21:552–556.

Jena VEB Gustav Fischer Verlag.

Wickham, H. 2016. [ggplot2: Elegant graphics for data analysis](#). Springer-Verlag New York.

Ye, J., G. Coulouris, I. Zaretskaya, I. Cutcutache, S. Rozen, and T. L. Madden. 2012. [Primer-BLAST: A tool to design target-specific primers for polymerase chain reaction](#).

BMC Bioinformatics 13:134.

Yu, G. 2020. [Using ggtree to visualize data on tree-like structures](#). Current Protocols in Bioinformatics 69:e96.

Yu, G., D. Smith, H. Zhu, Y. Guan, and T. T.-Y. Lam. 2017. [Ggtree: An r package for visualization and annotation of phylogenetic trees with their covariates and other associated data](#). Methods in Ecology and Evolution 8:28–36.

1
2
3 Zoological Nomenclature, I. C. on, W. D. L. Ride, I. T. for Zoological Nomenclature., I.
4
5 U. of Biological Sciences. General Assembly, and E. Natural History Museum (London.
6
7 1999. [International code of zoological nomenclature = code international de](#)
8
9 [nomenclature zoologique](#). London, International Trust for Zoological Nomenclature, c/o
10
11 Natural History Museum, 1999.
12
13
14

15 Zweifel, R. G. 2003. A new species of microhylid frog, genus oreophryne, from papua
16
17 new guinea. American Museum Novitates 1–8.
18
19
20

21 Zweifel, R. G. 1956. Microhylid frogs from new guinea, with descriptions of new species.
22
23 American Museum Novitates 1–49.
24
25

26 Zweifel, R. G., H. Cogger, and S. Richards. 2005. [Systematics of Microhylid Frogs,](#)
27
28 [Genus Oreophryne, Living at High Elevations in New Guinea](#). American Museum
29
30 Novitates 3495:1–25.
31
32
33

34 Zweifel, R., J. Menzies, and D. Price. 2003. [Systematics of microhylid frogs, genus](#)
35
36 [oreophryne, from the north coast region of new guinea](#). AMERICAN MUSEUM
37
38 NOVITATES 1–31.
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56

Table 1.
Locality information for samples sequenced in this study. See Hill, et al., 2023, Table 1 for additional metadata.

			Latitude
ID	Species	Locality	Latitude, Longitude
BJE01248	<i>Oreophryne frontifasciata</i>	Kabupaten Halmahera Timur, Halmahera Island, North Maluku	1.3667, 128.2330
BJE01143	<i>Oreophryne senckenbergiana</i>	Kabupaten Halmahera Utara, Halmahera Island, North Maluku	1.7899, 127.8910
BJE01144	<i>Oreophryne senckenbergiana</i>	Kabupaten Halmahera Utara, Halmahera Island, North Maluku	1.7899, 127.8910
BJE01120	<i>Oreophryne senckenbergiana</i>	Kabupaten Halmahera Utara, Halmahera Island, North Maluku	1.8228, 127.8300

Table 2.

Description of traits surveyed. Several traits are traditionally noted for Asterophryinae species: presence of two palatal folds, horizontal pupil, and lack of vomerine teeth (Boettger 1895; VanKampen 1923; Parker 1934).

Column*	Trait	Definition
1	Procoracoids	Size and shape of the procoracoid cartilages, CR= curved, reduced
2	Clavicles	Size and shape of the clavicles, CR= curved, reduced
3	Omasternum	Presence (+) or absence (-) of the omasternum
4	Sternum	Cartilaginous sternum or sternal plate
5	Call type	Qualitative description of the call type
6	Finger:toe width	The ratio of finger pad width to toe pad width
7	Pectoral ligament	Cartilaginous or ligamentous connection between scapula and procoracoid
8	Rel toe length	Relative length of 5th toe compared to length of 3rd toe (5<3, 5=3, 5>3)
9	Toe webbing	Toe webbing assessed on 5th toe (absent (-), basally webbed, well webbed)
10	Tympanum:eye diameter	Tympanum diameter relative to eye diameter ($\frac{1}{4}$, $\frac{1}{3}$, or $\frac{1}{2}$)
11	Reference	Citation
	Pupil horizontal	Whether the pupil is horizontal
	Teeth	Whether teeth are present
	Palatal folds	Whether two horizontal palatal folds in front of the pharynx are present

*Corresponding column in Table 3.

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46

Table 3.
Morphological character analysis.

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Genus	species	proc	clav	omo	ster	call type	fing: toe width	pec lig	rel toe len	toe web	tymp:eye diam	reference
<i>Oreophryne</i>	<i>anamiatoi</i>	CR	CR	–	cart	chuckle or cackle	1.2	cart	5>3	–	1/4	Kraus Allison 2009
	<i>anulata</i> ^l	CR	CR	–	cart		slightly larger		5>3	–	1/2	Stejneger 1908
	<i>aurora</i>	CR	CR	–	cart	harsh croaks		lig	5>3	webbed	1/4	Kraus 2016
	<i>biroi</i>	CR	CR	–	cart	rattle	1.1	lig	5=3	webbed	1/4	Zweifel 2003
	<i>brachypus</i>	CR	CR	–	cart	long squeak	1.1	lig	5>3	webbed	1/4	Zweifel 2003
	<i>cameroni</i>	CR	CR	–	cart	peep		cart	5>3	basal	1/4	Kraus 2013
	<i>frontifasciata</i> ^l	CR	CR	–	cart		slightly larger	cart	5>3	basal	1/2	Horst 1883, Brongersma 1948
	<i>geislerorum</i>	CR	CR	–	cart	rattle		lig	5>3	webbed	1/3	Zweifel 2003
	<i>graminis</i> ^l	CR	CR	–	cart	peep	approx equal	cart	5>3	–	1/3	Gunther Richards 2012
	<i>inornata</i>	CR	CR	–	cart	honk	1.0	lig	5>3	basal	1/4	Kraus 2016
	<i>lemur</i>	CR	CR	–	cart	honk		lig	5>3	basal	1/4	Kraus 2016
	<i>loriae</i>	CR	CR	–	cart	honk		lig	5>3	webbed	1/3	Kraus 2016

Table 3.

Morphological character analysis.

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Genus	species	proc	clav	omo	ster	call type	finger: toe width	pec lig	rel toe len	toe web	tymp:eye diam	reference
	<i>nana</i> ¹	CR	CR	–	cart		approx equal	cart	5<3	–	1/2	Brown Alcala, 1967
	<i>notata</i>	CR	CR	–	cart	peep		cart	5>3	–	1/4	Zweifel 2003
	<i>parkeri</i>	CR	CR	–	cart	peep		lig	5>3	webbed	1/3	Kraus 2013, Zweifel 2003
	<i>parkopanorum</i>	CR	CR	–	cart			cart	5<3	–	1/4	Kraus 2013
	<i>philosylleptoris</i>	CR	CR	–	cart	honk	1.0	lig	5>3	webbed	1/3	Kraus 2016
	<i>senckenbergiana</i> ¹	CR	CR	–	cart		slightly larger		5=3	basal	1/3	Boettger 1895, Parker 1934
<i>Auparoparo</i> ²	<i>brunnea</i> ¹	CR	CR	–	cart	rattle	larger	lig	5=3	basal	1/3	Kraus 2017
	<i>equus</i>	CR	CR	–	cart	whinny	1.3	lig	5=3	basal	1/4	Kraus 2016
	<i>ezra</i>	CR	CR	–	cart	chatter		lig	5>3	basal	1/4	Kraus Allison 2009
	<i>insulana</i>	CR	CR	–	cart	whinny		lig	5>3	basal	1/4	Kraus 2016, Zweifel 1956
	<i>matawan</i>	CR	CR	–	cart	rattle		lig	5≥3	–	1/4	Kraus 2016
	<i>meliades</i>	CR	CR	–	cart	whinny	1.2	lig	5=3	basal	1/3	Kraus 2016
	<i>penelopeia</i>	CR	CR	–	cart	whinny	1.3	lig	5>3	basal	1/4	Kraus 2016
	<i>phoebe</i>	CR	CR	–	cart	whinny	1.2	cart	5>3	webbed	1/4	Kraus 2017

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46

Table 3.
Morphological character analysis.

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Genus	species	proc	clav	omo	ster	call type	finger: toe width	pec lig	rel toe len	toe web	tymp:eye diam	reference
	<i>picticrus</i>	CR	CR	–	cart	whinny	1.3	lig	5>3	basal	1/3	Kraus 2016
	<i>variabilis</i> ¹	CR	CR	–	cart		much larger	cart		–	>2/3	Boulenger 1896

¹Character states obtained from the literature, specimens not available for examination in this study.

²Hill et al. 2022 places *Aphantophryne pansa* within *Auparoparo*, but the status of *Aphantophryne* requires further study with samples from additional species and localities.

*External characters were examined from specimens, internal characters and pupil shape taken from the literature.

Columns: (1) Procoracoids and (2) Clavicles CR= curved reduced, (3) Omasternum , (4) Sternum, (5) Call Type, (6) Finger to Toe Ratio, (7) Pectoral Connector Type, (8) Relative Toe Length of the fifth to third toe, (9) Toe Webbing, (10) Tympanum to Eye Diameter Ratio, (11) Reference. See Table 2 for character states.

Only